A3-PS015-TC031

Explainable Home and Last-Mile Connectivity Diagnosis

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AIORI Hackathon 3
October 13, 2026

Problem Description

Clear outline of the problem being addressed:

- Home internet issues can arise from gateway reachability, DNS, packet loss or congestion.
- The same symptom, such as slow loading, can have multiple possible causes.
- Users may receive technical errors without an understandable explanation.
- A diagnosis must be based on measurable evidence and should acknowledge uncertainty.

Project Aim: Provide an evidence-based diagnosis and explain why the system reached that result.

Background

Background necessary to understand the problem and the proposed optimization:

Home connectivity relies on several components, including the user's device, local network, gateway, DNS resolver and the upstream network path. A failure in one component may cause symptoms that resemble failures elsewhere.

The proposed tool will run controlled tests and interpret their results using transparent decision rules. It will examine:

- Reachability to the configured gateway
- DNS lookup response
- Latency and packet loss
- Performance changes under controlled congestion

A failed test is an indicator, not always proof of a root cause. The tool will report “Inconclusive” when evidence is insufficient.

Literature & RFC Review

Review of relevant standards, documentation and measurement tools

[1] ICMP (RFC 792)

Defines Internet Control Message Protocol messages used for network-layer feedback and diagnostics.

[2] DNS Concepts and Facilities (RFC 1034)

Describes the domain name system, including name servers and resolvers.

[3] RIPE Atlas Documentation

Describes a platform for internet measurements, including ping, DNS and traceroute.

Relevance: These standards and tools provide the measurements and protocol background. ConnectX will combine controlled tests with transparent rules and a human-readable explanation.

Proposed Solution

Detailed description of the overall solution.

The proposed system will perform controlled diagnostics and provide an explainable result.

- Run tests for gateway reachability, DNS response, latency and packet loss.
- Collect and organise the measurements with their test conditions.
- Apply explicit rules to identify patterns consistent with the four target scenarios.
- Show the likely cause together with the supporting evidence.
- Return “Inconclusive” when the available tests cannot isolate the cause.
- Avoid unsupported claims about ISP-internal faults or Wi-Fi interference without suitable evidence

Optimization Proposed

List of Optimizations and Enhancements

AREA	BASIC APPROACH	PROPOSED IMPROVEMENT
Testing	Run separate manual checks	Guide the user through a consistent test sequence
Results	Raw output may be difficult to interpret	Present findings in plain language with evidence
Decision	A single failure may be misleading	Compare multiple measurements before concluding
Uncertainty	Risk of forcing one diagnosis	Return “Inconclusive” if evidence is insufficient

Solution Architecture & Design

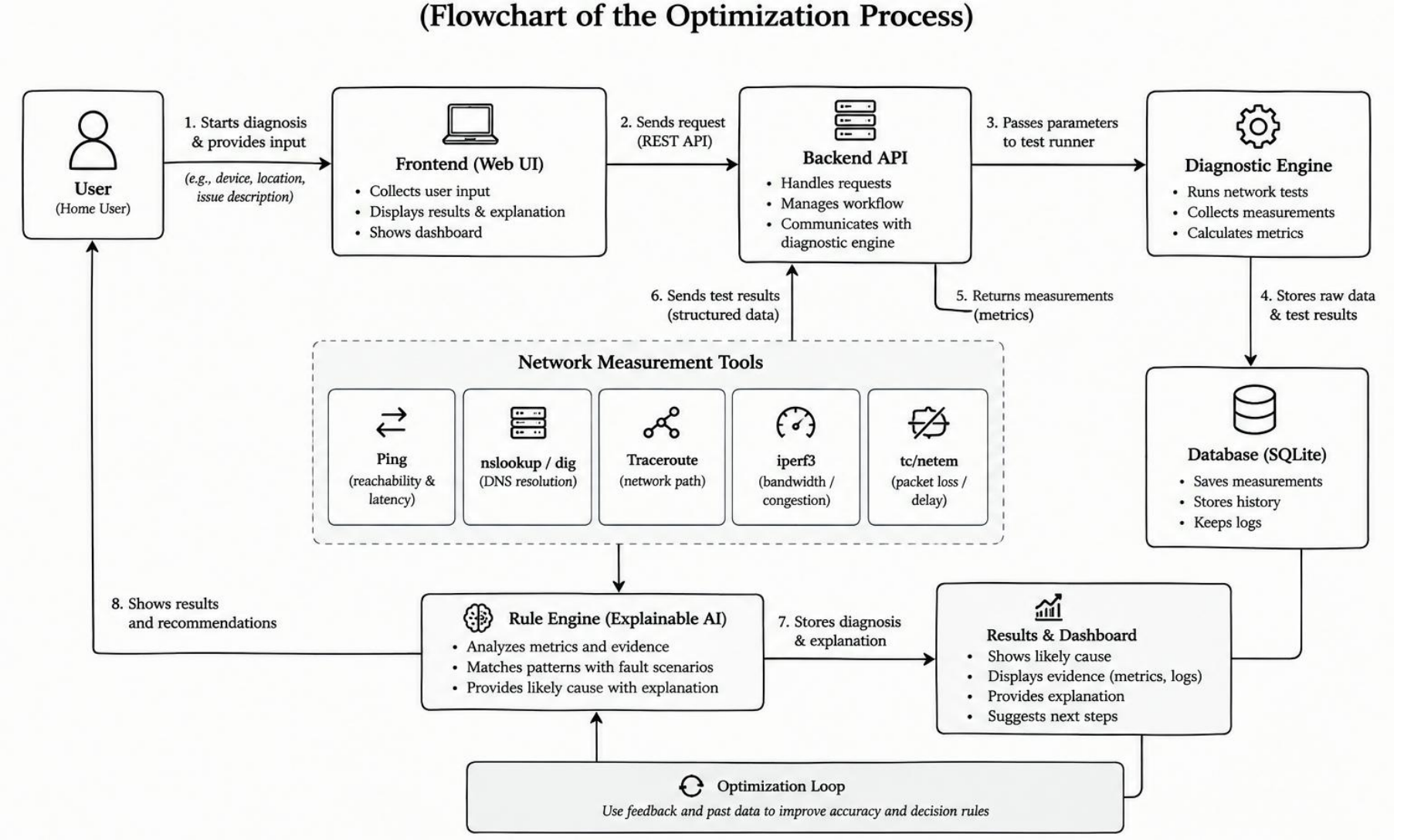


Figure: System Architecture UML Diagram

Data Flow Diagram

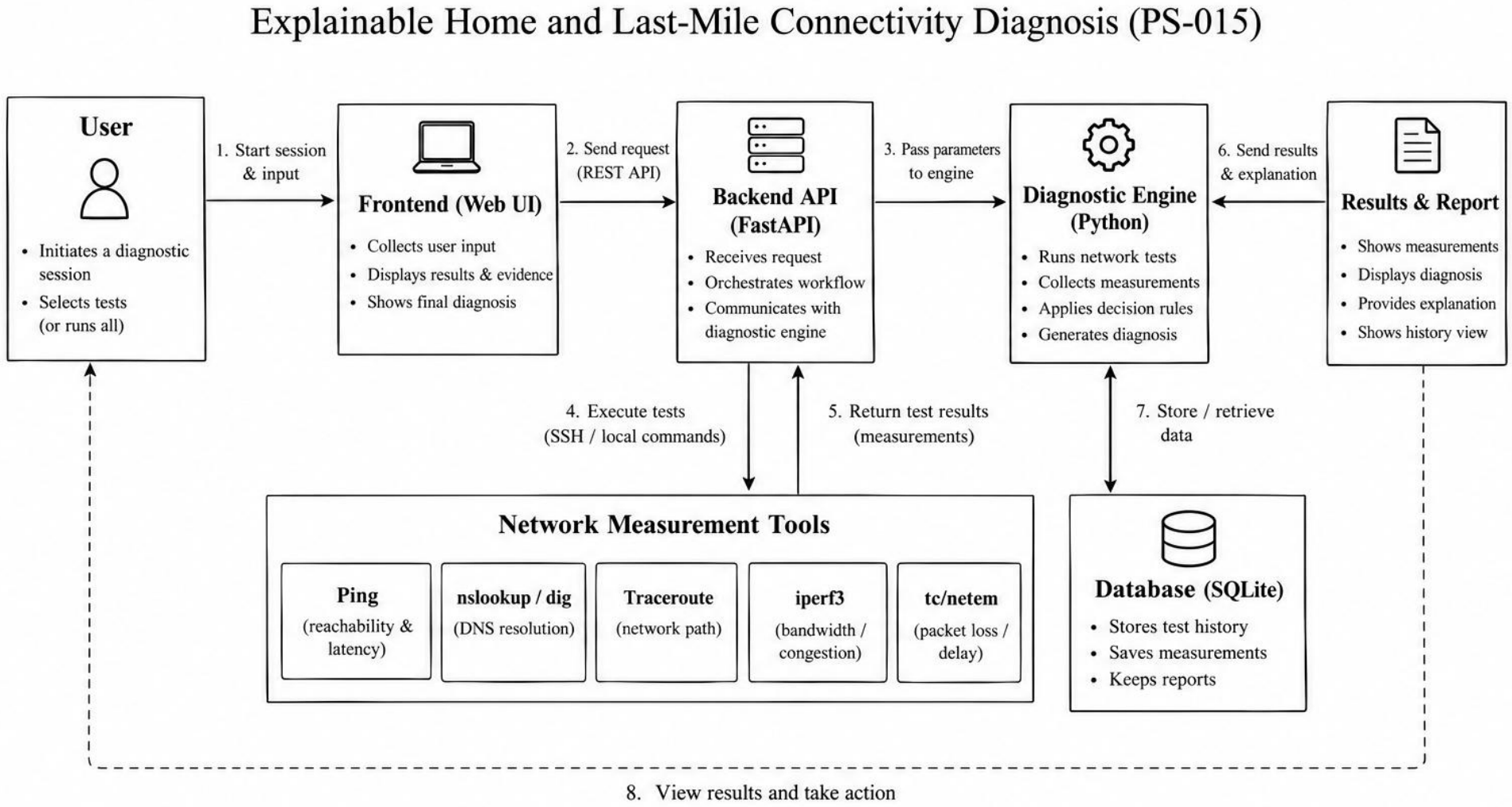


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Proposed milestones and methodology

- Phase 1 — Confirm requirements and prepare the development environment.
- Phase 2 — Set up the controlled network lab and implement the four fault scenarios.
- Phase 3 — Build the diagnostic tests and evidence collection.
- Phase 4 — Implement the rule engine, backend API and dashboard.
- Phase 5 — Test normal conditions, each fault scenario and inconclusive cases.
- Phase 6 — Prepare documentation, demonstration and final presentation.

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[1] J. Postel, “Internet Control Message Protocol,” RFC 792, Internet Engineering Task Force, 1981.
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Thank You